6 **(a)** Prove that $\dfrac{\sin\theta}{1-\cos\theta} - \dfrac{1}{\sin\theta} = \cot\theta$. **[4]**

(b) Hence find the exact roots of the equation $\dfrac{\sin\theta}{1-\cos\theta} - \dfrac{1}{\sin\theta} = 3\tan\theta$ in the interval $0 \leqslant \theta \leqslant \pi$. **[3]**